

ML Specification Report — siim-isic-melanoma-classification

Melanoma Classification from Dermoscopy Images + Patient Metadata • our from-scratch agent (ViT/CNN fine-tunes + metadata GBDT, convex rank blend — deep-learning fallback lane)

Figures grounded in the competition's *facts.json* (built by *benchmark_infra/collect_dl_facts.py* from *config.yaml*, *experiments.json*, *eda_summary.json*, *discovery.json*, *res_*.json*, *meta_results.json*, *blend_results.json*, *dossier.json*, *STATUS.md*, *headless_run2.log*, *scripts/vision.py*), with grading figures from the offline MLE-bench records in *benchmark_results/run2/*. This is a special (deep-learning fallback) lane: grading is offline MLE-bench against a held-out private split — there is no live Kaggle leaderboard, and the framework's DL fields (backbone, learning-rate schedule, batch size, epochs, augmentation, transfer learning) are native here rather than translated from GBDT.

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Overview

The task is to predict the probability that a skin-lesion image is malignant melanoma, from the dermoscopy image plus patient-level metadata (age, sex, anatomical site); the metric is **ROC-AUC (maximize)** on a heavily imbalanced label (~1.8% positive). Image classification is outside the GBDT tree-search pipeline, so the agent ran the vision fallback: cheap metadata/image-statistic models, a discovery LR sweep, five full backbone fine-tunes, and a convex rank blend. The champion is a **6-member convex rank blend — vit_small at 384/256/320 px + efficientnet_b0 + CatBoost/LightGBM on metadata & image statistics** — scoring **honest leave-fold-out CV AUC 0.93214** (optimistic full-OOF 0.93291; patient-grouped cross-check 0.93278; best solo vit_small @384 px 0.92596) and grading **0.93529** on the offline MLE-bench private split — **above median (0.9128) but short of bronze (0.937)**:

no medal. The NVIDIA reproduce-agent grades 0.89822 and AIDE 0.88579 — **our agent wins the three-way comparison.**

Why it matters. Melanoma causes the large majority of skin-cancer deaths, yet is highly curable when caught early; dermatologist-grade triage from a dermoscopy photo plus context (the clinical "ugly duckling" sign — a lesion suspicious relative to the patient's other moles) is exactly what this model family automates.

Data

Purpose of Data. Each row is one lesion image (`image_name`) with patient metadata; predict `target` , the malignancy probability — binary classification scored as ranking. **Data Format** is **JPEG/DICOM images + a tabular CSV** of metadata. **Data Volume: 28,984 training images / 4,142 test images** (8 vs 5 metadata columns; `benign_malignant` and `diagnosis` are train-only), re-split by mle-bench from the original 33,126-image train set. **513 positives** in train — a positive rate of 0.0177.

Data Quality. Metadata missingness is small: fractions 0.0019 (`sex`), 0.0020 (`age_approx`), 0.0161 (`anatom_site_general_challenge`). Images span 81 distinct resolutions in train / 28 in test, with a 0.0017 fraction of test rows at an unseen resolution. Patients: 2,056 in train / 1,457 in test, with images-per-patient mean 14.1 (max 104). **The defining wrinkle is the split's structure:** the run's EDA refuted the dossier's patient-disjoint hypothesis — **all 1,457 test patients also appear in train** (overlap 1457/1457; 599 patients are train-only), with per-patient test share mean 0.124 / median 0.111 vs an overall test share of 0.125. mle-bench re-split the images **randomly by image**, so the graded test set is iid-by-image, not patient-disjoint like the real SIIM competition.

Strongest recorded EDA/modelling signals:

- **Cheap image colour/texture statistics are the largest tabular lever:** removing them drops the LightGBM metadata model 0.88574 → 0.82114 (−0.06460 AUC); removing resolution features −0.01417; removing patient-relative features −0.01330.
- **The ugly-duckling sign appears directly in the model:** patient-relative `ud_*` z-scores occupy 8 of the top-20 LightGBM importances.
- **Class imbalance needs no reweighting** — AUC is a pure ranking metric, so all members train without class weighting.

Annotation Guidelines. The label is `target` $\in \{0, 1\}$ (malignant or not); a submission is a real-valued malignancy score per image, scored by ROC-AUC — raw scores, no threshold or calibration.

Feature Set. The vision members consume raw pixels; the metadata members consume **54 features:** sex/age/site, image resolution, prep-time colour/texture statistics (central-disc "lesion" vs outer-ring "skin" colour, contrast, gradient energy, vignetting), and patient-relative ugly-duckling z-scores.

Splitting strategy. Primary CV is **image-level StratifiedKFold(5, seed=42)** — deliberately, because the graded split is iid-by-image; a patient-grouped `StratifiedGroupKFold` would measure a strictly harder problem than the one being scored. Every member additionally reports a **patient-grouped score as a leakage detector**: the two agreed everywhere, largest gap 0.0005 AUC across all 10 members (`lr_allfeat`, 0.85467 image-level vs 0.85418 patient-grouped) — no member exploits patient identity. Folds are persisted to `cache/r2_folds.csv` and shared by every model, so all scores are digit-for-digit comparable. **LB status**: offline MLE-bench grade only — no live public/private leaderboard.

Models & Architecture

Purpose of Architecture. Maximize ROC-AUC for malignancy ranking. **Architecture Type** is a **heterogeneous ensemble of fine-tuned ImageNet-pretrained backbones plus metadata GBDTs, combined by a convex rank blend** — the DL fields are native. The champion carries 6 non-zero members:

Member	Backbone / model	px	Seed	Solo CV AUC	Blend weight
vit_small_384	vit_small_patch16_224	384	42	0.92596	0.388
vit_small_256	vit_small_patch16_224	256	42	0.92309	0.286
vit_small_320	vit_small_patch16_224	320	2024	0.92024	0.102
lgb_meta	LightGBM, 54 metadata features	—	42	0.88574	0.102
cat_meta	CatBoost, 54 metadata features	—	42	0.89171	0.082
effb0_256	efficientnet_b0	256	42	0.88447	0.041

Zero-weighted candidates: `resnet50_256` (solo 0.89166), `xgb_meta` (0.88668), `lr_allfeat` (0.85467), and a level-2 LightGBM stack (0.92566).

Input Format / Dimension. Vision members take square RGB crops at 384 / 256 / 320 px (uint8 arrays preloaded to GPU; resize/augment/normalize on-GPU in batch); metadata members take a 54-dimensional feature vector.

Architecture Description. Each vision member is a `timm` backbone with `pretrained=True`, a 1-logit head, and drop-path 0.1 (`scripts/vision.py`). The blend layer **rank-transforms each member's OOF predictions, then runs Caruana greedy selection with replacement**; the reported champion score is the **honest leave-fold-out** estimate (weights refit on 4 folds, scored on the held-out fold). **Model Complexity**: parameter counts were **not recorded**; the members are fully identified by their published backbone names. A recorded structural finding: **input-resolution diversity within the ViT family beat architecture diversity** — the three `vit_small` scales take 0.78 of the total weight while both extra CNN families end at or near zero.

Training procedures

Training was a **staged discovery-first ladder** (tree search deliberately skipped — see below):

Stage	Configuration	AUC
1	metadata + image statistics, best tabular (CatBoost, 54 features)	0.89171
2	discovery sweep: 18 backbone×LR configs, fold 0, 2 epochs, 256 px, 52 min	best probe 0.8853
3	five full 5-fold fine-tunes, best solo (vit_small @384)	0.92596
4	convex rank blend, honest leave-fold-out	0.93214

The discovery sweep is the native learning-rate selection step (per-backbone LR mini-sweep, mandated by the vision experience library):

Backbone	1e-4	3e-4	1e-3	best LR
vit_small_patch16_224	0.8853	0.8686	0.8173	1e-4
efficientnet_b0	0.7686	0.8231	0.8782	1e-3
efficientnet_b3	0.7868	0.8489	0.8716	1e-3
resnet50	0.7608	0.8393	0.8694	1e-3
resnet18	0.7936	0.8672	0.8551	3e-4
convnext_tiny	0.8643	0.8468	0.7775	1e-4

CNNs peak at 1e-3, ViT/ConvNeXt at 1e-4 — the fifth domain confirming this pattern. The full fine-tunes:

Tag	px	Epochs	Batch	LR	CV AUC	patient-grouped	Wall
vit_small_384	384	8	64	1e-4	0.92596	0.92589	5345.5 s
vit_small_256	256	10	96	1e-4	0.92309	0.92295	2942.0 s
vit_small_320	320	9	80	1e-4	0.92024	0.92007	4133.1 s
resnet50_256	256	10	96	1e-3	0.89166	0.89179	4218.9 s
effb0_256	256	10	96	1e-3	0.88447	0.88440	2883.2 s

The **Loss Function** is **binary cross-entropy with logits** (`scripts/vision.py`), with model selection on ROC-AUC and **no class weighting** (a ranking metric gains nothing from resampling a 0.0177 positive rate). The **Optimization Algorithm** is **AdamW** (weight decay 1e-4). The **Learning Rate** (native) is backbone-specific per the discovery sweep — 1e-4 for ViT members, 1e-3 for the CNNs — under a native **Learning Rate Scheduler: linear warmup over the first 10% of steps, then cosine decay** (per-step, `scripts/vision.py`). The **Batch Size** (native) is 64–96 depending on member (table above), for **8–10 epochs**, under **bf16 autocast**.

Data Augmentation (native DL field): **dihedral-group augmentation** ("medium" setting — random h-flip, v-flip, and transpose, all label-preserving since dermoscopy is orientation-free), plus **4× flip TTA** at

prediction time. One honestly recorded gap: dihedral augmentation was adopted but never ablated against no-aug, so its contribution is unmeasured. **Transfer Learning** (native DL field): **ImageNet-pretrained timm backbones** for every vision member — allowed by the rules (`pretrained_models_allowed: true`) and named by the dossier as the run's main source of injected prior knowledge.

Training Duration: ~6.5 h for the whole run, all in the foreground — 52 min discovery plus fine-tune walls of 48–89 min each (per-member seconds in the table). **Training Memory: not recorded.**

Reproducibility Standards. `SEED=42` throughout (`vit_small_320` deliberately uses 2024 for decorrelation); folds persisted to `cache/r2_folds.csv` and shared by every member; `torch.manual_seed` / `cuda.manual_seed_all` / `np.random.seed` set per fold. `cuda.benchmark` is on — a deliberate speed trade whose recorded consequence is that CNN OOFs reproduce to ~1e-3 AUC, not bitwise. **Tree search was deliberately skipped:** a solo node is a 5-fold fine-tune costing 48–89 min, so the prescribed 60-node budget is 50–90 GPU-hours against ~8 h. Blend-layer-only tree search was considered and rejected as redundant — the greedy-with-replacement search already covers that space, and three structurally different member sets land within 0.00001 of each other (0.93204 / 0.93204 / 0.93203), i.e. the blend is saturated.

Recorded negative results: the **level-2 LightGBM stack lost to the convex blend** (0.92566 vs 0.93214, same direction as the s3e14 Ridge-stacking counterexample); **extra CNN families bought nothing** (resnet50 zero-weighted despite a respectable 0.89166 solo); and the **discovery-scale gap widened rather than shrank** (`vit_small` led `efficientnet_b0` by 0.007 at 2 epochs and 0.039 at 10 epochs), a direct counterexample to the digit-recognizer prior "don't over-prune on discovery scores".

Inference procedures

Decision Threshold is N/A — ROC-AUC is a pure ranking metric; raw malignancy scores are submitted with no threshold or calibration. Inference-time post-processing: **4× flip TTA** per vision member, then **rank-transform and convex-weighted averaging** of the 6 members' test predictions. The submission has 4,142 rows with id order verified against `sample_submission.csv`. **Inference Duration / Inference Memory: not recorded.**

Evaluation & Benchmarking

Task for Performance Evaluation. Rank test lesions by malignancy probability. **Performance Metrics.** ROC-AUC (maximize): metadata 0.89171 → best solo fine-tune 0.92596 → blend **0.93214** honest leave-fold-out (the single largest gain in the run, +0.037 AUC, came from training a strong backbone at all); offline MLE-bench grade **0.93529** against thresholds gold 0.9455 / silver 0.9401 / bronze 0.937 / median 0.9128 — **above median, no medal** (0.93529 falls short of bronze 0.937). The grade sits slightly above

the honest CV (0.93214) and close to the optimistic full-OOF (0.93291), consistent with a well-calibrated validation.

Performance Benchmarking. Grading is **offline MLE-bench** (all three agents scored on the same held-out private split — not a live leaderboard):

Agent	Approach	ROC-AUC (offline grade)	Medal
Our agent	from-scratch ViT/CNN fine-tunes + metadata GBDT, convex rank blend	0.93529	none (above median)
NVIDIA	(approach not recorded in the grade record)	0.89822	none (below median)
AIDE	(approach not recorded in the grade record)	0.88579	none (below median)

Our agent is the only one above the median threshold, with a comfortable margin over both comparators. The honest caveats: nobody medalled, the comparison is grade-only (the comparators' pipelines are not described in the run2 grade records), and our margin traces to the discovery-first backbone selection plus resolution-diverse ViT blending rather than any exotic component.

Fields marked "not recorded": model parameter counts; training peak memory; inference duration and memory; the contribution of dihedral augmentation (adopted but never ablated); NVIDIA/AIDE approach details. Grading is offline MLE-bench — no public/private live leaderboard exists for this lane.